A3-PS003-TC198

Federated Traffic-Flow Prediction with Communication-Cost Evaluation

Category: Smart Cities

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PS-003

Problem Description

Clear outline of the problem being addressed.

- Centralized traffic forecasting can require roadside sensors or edge clients to continuously send raw traffic data to a central server.
- Continuous transmission increases communication overhead and centralizes sensitive mobility information.
- Traffic patterns vary across locations and regions, creating non-IID client distributions that can affect federated convergence and prediction quality.
- A meaningful evaluation must compare centralized and FedAvg-based forecasting using MAE/RMSE together with communication bytes per training round and convergence.
- The project therefore studies whether federated traffic-speed forecasting can retain useful prediction quality while reducing unnecessary communication on a public historical traffic dataset.

Background

Traffic forecasting depends on both road-network structure and time-varying sensor behavior.

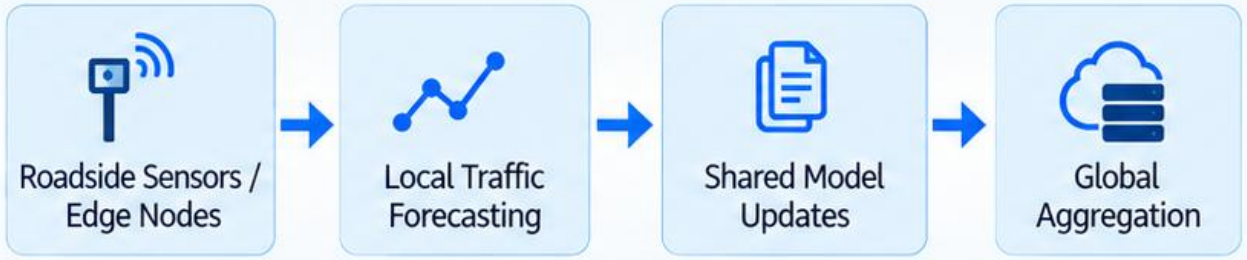
Key Context

- **Centralized forecasting sends continuous sensor data** to a cloud server, increasing bandwidth use and concentrating travel logs.
- **Traffic is naturally spatio-temporal**: nearby roads influence each other, and speed changes over time.
- **Public METR-LA data enables realistic evaluation** of forecasting quality and communication trade-offs.

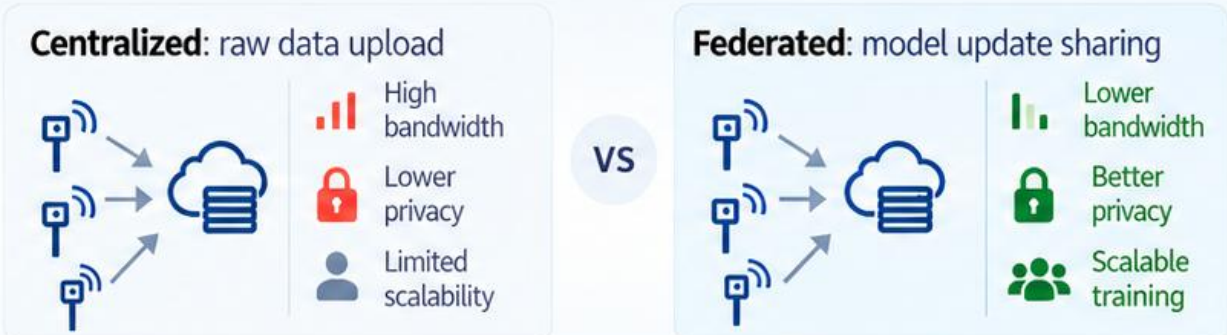
Why optimization is needed

- Lower communication cost
- Better privacy posture
- Scalable distributed training

Federated Traffic-Flow Forecasting Pipeline



Centralized vs. Federated Approach



Literature & RFC Review

A detailed study of the most relevant existing proposals, studies and standards for federated, graph-based traffic forecasting and communication-efficient learning.

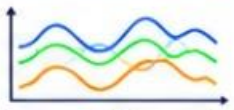
#	Year	Paper / Model	Key Contribution	Relevance to Our Project
1	2017	**FedAvg** (McMahan et al.)	Introduced federated learning with iterative model averaging across multiple clients, handling non-IID data.	Core federated optimization method for our multi-client traffic forecasting setup. Reduces data sharing and communication overhead.
2	2018	**DCRNN** (Li et al., ICLR 2018)	Spatio-temporal traffic forecasting on directed graphs using diffusion convolution and recurrent neural networks.	Key reference for graph-based traffic forecasting. Widely evaluated on the METR-LA dataset, which we also use (traffic speed).
3	2018	**STGCN** (Yu, Yin & Zhu, IJCAI 2018)	Graph convolutional networks for spatio-temporal modeling, capturing spatial and temporal dependencies in traffic data.	Demonstrates the effectiveness of graph-based spatio-temporal models and motivates our Graph + LSTM design.
4	2020	**Communication-Efficient FL** (Sattler et al., IEEE TCOM 2020)	Selective client participation, model compression and reduced communication for federated learning.	Directly relevant to our study of communication cost vs. accuracy trade-off, where only a subset of regional clients participate in each round.

How these works connect to our project



Graph-based traffic modeling (DCRNN, STGCN)

→



Temporal forecasting (LSTM-based models) (DCRNN, STGCN)

→



Federated learning across regional clients (FedAvg)

→



Communication-efficient training & evaluation (Sattler et al.)

Key relevance to this project

1. Graph + LSTM multi-horizon traffic-speed forecasting using real METR-LA dataset.
2. 4 regional federated clients with non-IID data and selective client participation.
3. Joint evaluation of accuracy, convergence and communication cost.

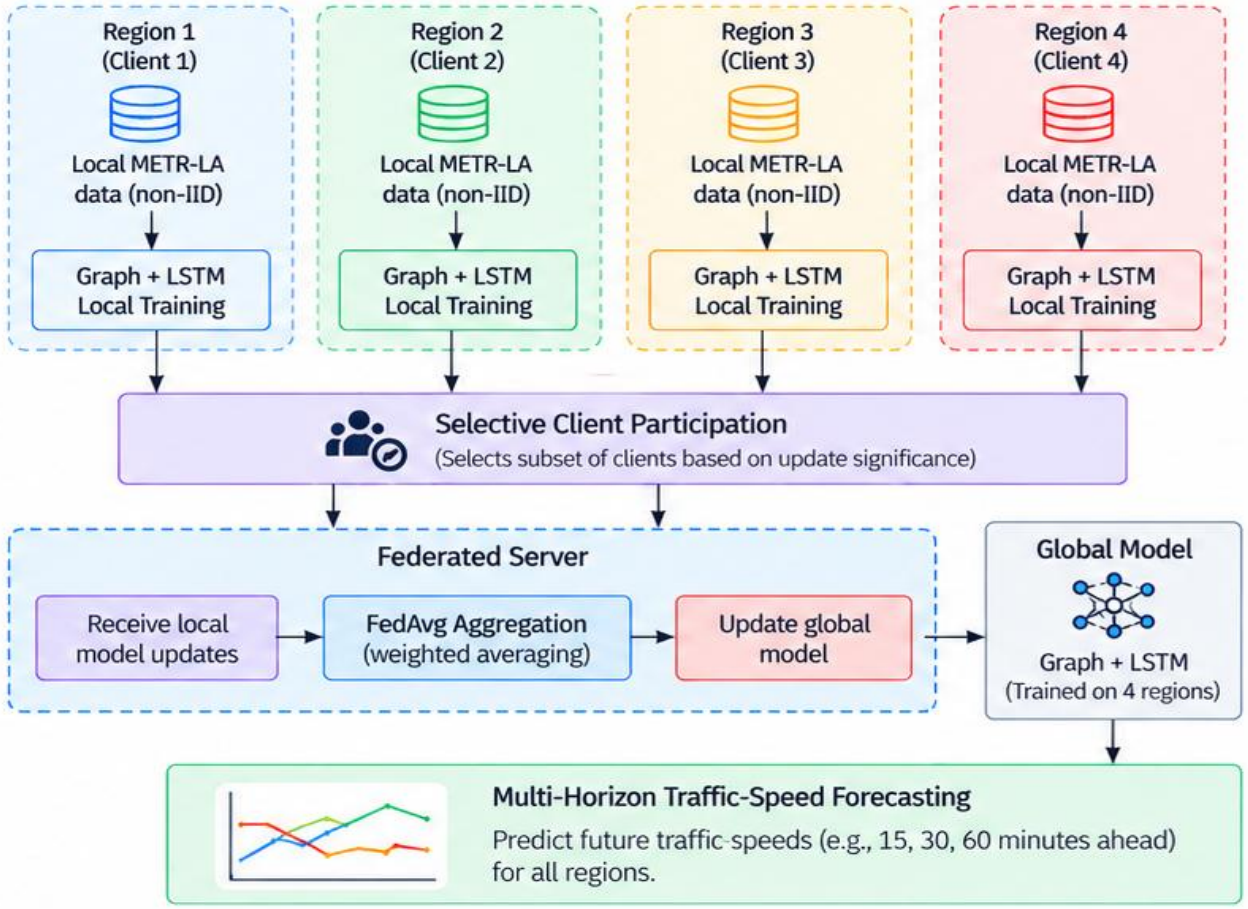
Proposed Solution

Detailed description of the overall solution.

We propose a federated, graph-based deep learning framework for multi-horizon traffic-speed forecasting using real-world METR-LA data. The road network is modeled as a graph where nodes represent sensors and edges capture spatial dependencies (e.g., road connectivity). Each regional client trains a local model on its own non-IID data, enabling privacy-preserving learning without sharing raw data.

The prediction model combines graph convolutional layers to learn spatial features from the road network with LSTM layers to capture temporal dependencies in traffic speeds. A federated learning setup with FedAvg aggregates local model updates at a central server. To reduce communication cost, we use a selective client participation strategy, where only a subset of clients (those with significant updates) participate in each round.

The trained global model provides multi-horizon traffic-speed predictions for all regions while balancing accuracy, convergence, and communication efficiency.



1 Data & Graph Construction
- Preprocess METR-LA speed data
- Build road network graph (adjacency matrix)

2 Local Model Training
- Graph Convolution + LSTM model
- Train on regional non-IID data

3 Selective Participation
- Choose subset of clients based on update significance
- Reduces communication cost

4 Model Aggregation
- FedAvg with weighted averaging
- Update global model

5 Forecasting
- Multi-horizon traffic-speed prediction
- Evaluate accuracy, convergence and communication cost

Optimization Proposed

Targeted optimizations improve forecasting quality while reducing communication overhead in federated traffic forecasting.

Before vs After Optimization

Optimization Area	Before	After	Benefit
Data Transfer	Continuous raw data upload to central cloud	Local training with parameter updates only	Lower communication cost and reduced raw-data dependence
Spatial Modeling	Mainly temporal or isolated sensor view	Graph + LSTM uses road connectivity and temporal patterns	Better representation of traffic dynamics
Client Participation	Every client contributes in every round	Selective client participation based on update significance	Fewer bytes per round with efficient training
Regional Data Handling	No explicit heterogeneity study	IID and non-IID client partitions across regions	Realistic federated evaluation
Evaluation Strategy	Accuracy checked alone	MAE, RMSE, convergence, and communication bytes tracked together	Balanced quality-cost assessment
Forecasting Output	Limited prediction perspective	Multi-horizon forecasts at +5, +15, +30, +60 min	More useful operational insight

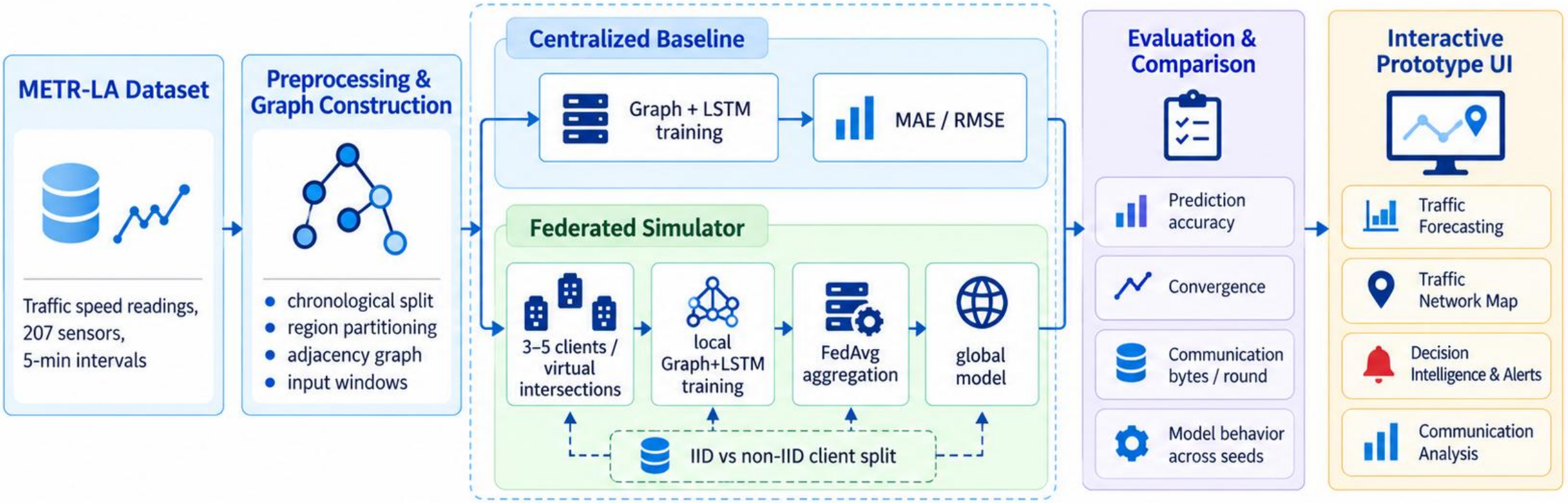
Optimization Pipeline



Expected Impact

- Lower communication load
- Better robustness on non-IID traffic data
- Scalable edge-friendly forecasting

Solution Architecture & Design



Component Legend

Data storage (database) · Server (computation) · Data flow (process flow) · Model update flow (federated learning) · Graph model (graph structure)

Key Features in Architecture

1. Spatiotemporal modeling with Graph + LSTM
2. Federated learning with FedAvg aggregation
3. Support for IID and non-IID client data split
4. Comprehensive evaluation and comparison
5. Interactive prototype for real-world visualization

Technology Used

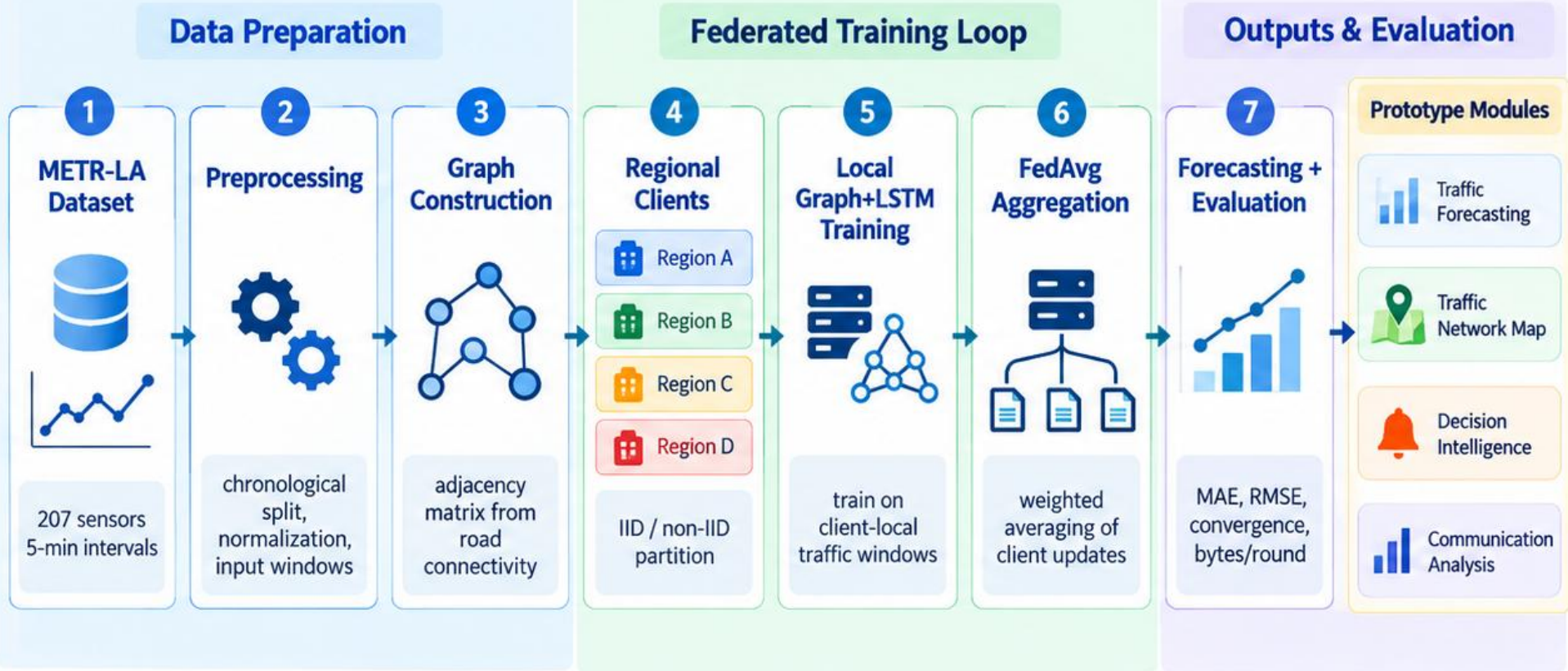
Python · PyTorch · Graph + LSTM · FedAvg · React / Frontend · Map UI

Data Flow Diagram

Overview of the architecture and end-to-end data movement in the federated traffic-forecasting pipeline.

Flow Summary

1. METR-LA speed readings are converted into graph-aware time windows.
2. Data is split across regional clients to simulate non-IID intersections.
3. Each client trains a local Graph + LSTM model without sharing raw sensor data.
4. Model updates are aggregated centrally and evaluated for accuracy and communication cost.



Deployment Timeline

Phased roadmap for building and validating our federated traffic-flow prediction system.



Bibliography

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  https://proceedings.mlr.press/v54/mcmahan17a.html
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Thank You for Your Attention!